

Towards Accelerating Transportation Research: Measuring the Practice of Open Science

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Abstract

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Keywords: Open Science, Large Language Models

1. Introduction

I need a citation to keep from getting a latex error, so [Nosek et al. \(2015\)](#) is a good one.

2. Analysis

2.1. Descriptive statistics and bivariate tests

The analysis dataset contains 10 480 papers that report quantitative studies (`is_quantitative_study`) among the full-length articles. Table 1 provides descriptive statistics for the dataset, organized by code availability. provides descriptive statistics for these papers, organized by code availability. The dataset contains 528 papers with code available, a little more than 0 % of the dataset. x In total, about 68.7% of the papers cite available data, but only 30.2% of the papers include a data repository; the remaining 1% neither cite data nor include a repository.

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Table 1: Descriptive Statistics of Data Set

		Code is not available (N=9952)		Code is available (N=528)	
		Mean	Std. Dev.	Mean	Std. Dev.
Open science score		2.2	0.6	3.4	1.1
Age of paper [years]		3.4	1.7	3.1	1.7
N. tables in paper		6.4	4.5	6.3	4.6
N. figures in paper		8.6	6.1	11.0	6.5
N. references		58.2	26.9	57.3	21.0
N. authors		3.8	1.7	3.8	1.7
Times cited		28.4	43.2	27.2	42.9
Review time [days]		254.3	148.9	268.6	141.4
Page count		18.4	6.3	19.8	6.6
		N	Pct.	N	Pct.
Open Access	Not open access	7346	73.8	324	61.4
	Open access	2606	26.2	204	38.6
Data availability	No repository or citation	7063	71.0	139	26.3
	Data repository is available	65	0.7	199	37.7
	Data is cited or linked	2810	28.2	96	18.2
	Data repository and citation	14	0.1	94	17.8
Journal	Interdisciplinary Perspectives	945	9.5	36	6.8
	Part A: Policy and Practice	1555	15.6	64	12.1
	Part B: Methodological	771	7.7	67	12.7
	Part C: Emerging Technologies	1797	18.1	212	40.2
	Part D: Transport and Environment	1911	19.2	72	13.6
	Part E: Logistics and Transportation Review	1534	15.4	41	7.8
	Part F: Traffic Psychology and Behaviour	1439	14.5	36	6.8
Region of corr. author	Africa	69	0.7	5	0.9
	Asia	4398	44.2	118	22.3
	Europe	2699	27.1	206	39.0
	North America	2074	20.8	156	29.5
	Oceania	521	5.2	27	5.1
	South America	191	1.9	16	3.0
Availability statement	Availability statement is not present	7062	71.0	139	26.3
	Availability statement is present	2890	29.0	389	73.7
Paper topic	Air & Freight Logistics	77	0.8	2	0.4
	Automated Driving & Human Factors	957	9.6	34	6.4
	COVID-19 Impact on Travel & Activities	434	4.4	17	3.2
	Data-Driven Modeling & Prediction	772	7.8	142	26.9
	Driver Behavior & Safety Risk	631	6.3	10	1.9
	Electric Vehicles & Ride-Sharing	372	3.7	29	5.5
	Optimization & Routing Algorithms	1513	15.2	114	21.6
	Public Transit Service & Demand	489	4.9	29	5.5
	Road Infrastructure & Emergency Management	49	0.5	1	0.2
	Social & Policy Aspects of Mobility	862	8.7	26	4.9
	Supply Chain & Market Strategies	775	7.8	6	1.1
	Traffic Flow & Network Control	710	7.1	45	8.5
	Transportation Emissions & Policy	848	8.5	20	3.8
	Travel & Mode Choice Behavior	883	8.9	28	5.3
	Urban Environment & Active Transport	580	5.8	25	4.7
Links to code repositories	Link to GitHub	4	0.0	329	62.3
	Link to other service	5	0.1	118	22.3
	No links	9943	99.9	81	15.3

We conducted a series of bivariate statistical tests on the dependent variables — specifically, whether the papers provide code or data — using t -tests on differences of means when the independent variable is numeric, and Pearson χ^2 tests of independence when the independent variable is categorical. The results of these tests revealed several statistically significant relationships that guided the analysis that follows in this section. To begin, papers that make code available are also more likely to share data. Newer papers make code available more frequently, and newer papers also tend to share data directly rather than simply cite existing data. Papers with corresponding authors in Europe and North America are more likely to make data available than papers with authors in Asia; the same trend holds for sharing code, with South American authors joining the group more likely to share than Asian authors. In absolute terms, authors based in Asia still contribute a substantial number ($N = 118$) of code repositories, but their share appears lower when expressed as a percentage of the total number of publications from the region ($N = 4516$). The usage of data varies widely by topic, with the topics “Urban Environment & Active Transport” and “COVID-19 Impact on Travel & Activities” having higher than expected rates of citing data from others (about 45% of papers cited or linked to external datasets in these topic categories). Papers in the topic “Data-driven modeling and prediction” share code at a much higher rate than other topics (about 15% of papers have code available). Other comparisons did not result in bivariate test statistics sufficient to reject a null hypothesis of no relationship or no difference in means. Note that the numbers reported in the analysis section are subject to estimation error arising from the methodology.

These findings are informative, but they may be the result of correlated omitted variables. A complete list of statistics of all variables can be viewed [here](#). In the next section we provide a choice model of data and code availability that can accommodate the simultaneous influence of multiple variables.

2.2. Incentives for availability

Open science techniques often require additional investment and effort by researchers. If the discipline does not require these efforts, researchers may respond to various incentives. In our data, two incentives that we capture may include a higher rate of citations on the resulting publication or a reduced review time prior to publication. Unfortunately, neither incentive is observable in the papers we studied.

In terms of the number of citations, Figures ?? and shows the citations per year — calculated as the number of citations in SCOPUS divided by the whole number of years since publication — for each paper ordered by paper acceptance date in SCOPUS. Figure ?? groups the papers by their data availability in the categories described above, and Figure ?? groups the papers by code availability. Both figures show a LOESS moving average rate of the mean citation rate per year, along with a 95% confidence interval of the mean. In neither case are the average citation rates different based on the availability of data or code. Figure ?? shows a temporary increase in the citation rate for papers with a data repository and citation in 2020 and 2021,

to the point where the uncertainty band does not overlap the uncertainty band of the other papers with different data sharing practices. Further examination of this effect showed that two papers with unusually high citations drive this effect.

```
lm_citations_ctl <- lm(log(citations_per_year + 1) ~ journal + lda_topic + region_normalized, data = p
lm_citations <- update(lm_citations_ctl, . ~ . + is_code_available + data_multi)

lm_reviewtime_ctl <- update(lm_citations_ctl, log(review_time_days + 1) ~ .)
lm_reviewtime <- update(lm_citations, log(review_time_days + 1) ~ .)

acitations <- anova(lm_citations_ctl, lm_citations)
areviewtime <- anova(lm_reviewtime_ctl, lm_reviewtime)

car::vif(lm_citations)
```

	GVIF	Df	GVIF ^{1/(2*Df)}
journal	7.644927	6	1.184717
lda_topic	8.023916	14	1.077208
region_normalized	1.262443	5	1.023579
is_code_available	1.766925	1	1.329257
data_multi	1.932137	3	1.116023

```
cm <- c(
  "(Intercept)" = "Intercept",
  "is_code_availableCode is available" = "Code is available",
  "data_multiData repository is available" = "Data repository is available (NC, R)",
  "data_multiData is cited or linked" = "Data is cited or linked (C, NR)",
  "data_multiData repository and citation" = "Data repository and citation (C, R)",
  "journalTransportation Research Part A: Policy and Practice" = "Journal: TR - A",
  "journalTransportation Research Part B: Methodological" = "Journal: TR - B",
  "journalTransportation Research Part C: Emerging Technologies" = "Journal: TR - C",
  "journalTransportation Research Part D: Transport and Environment" = "Journal: TR - D",
  "journalTransportation Research Part E: Logistics and Transportation Review" = "Journal: TR - E",
  "journalTransportation Research Part F: Traffic Psychology and Behaviour" = "Journal: TR - F",

  "lda_topicCOVID-19 Impact on Travel & Activities" = "Topic: COVID-19 Impact on Travel & Activities",
```

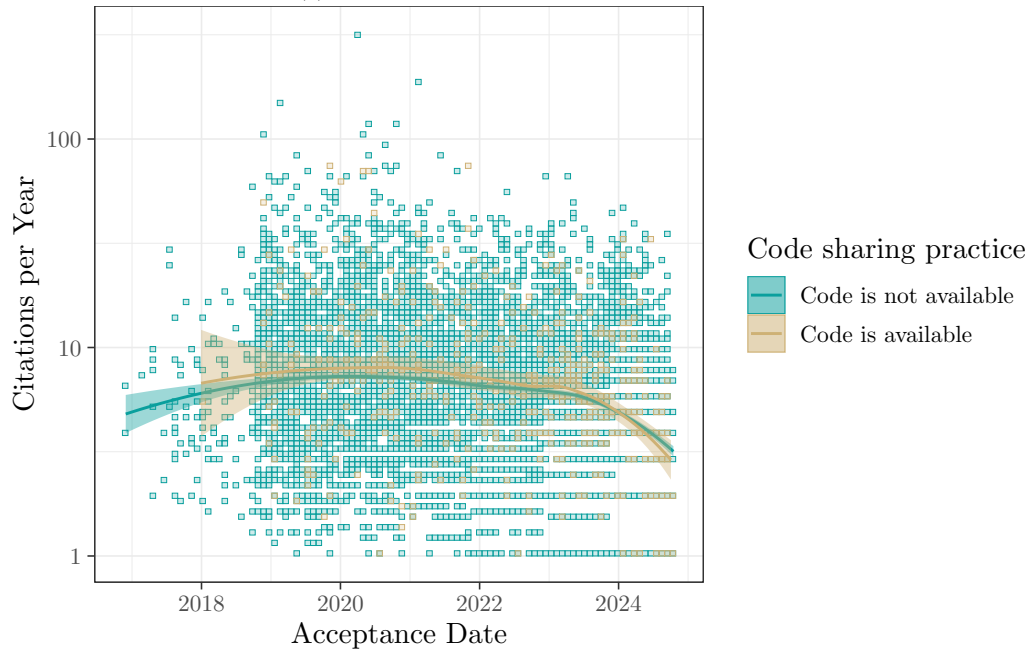
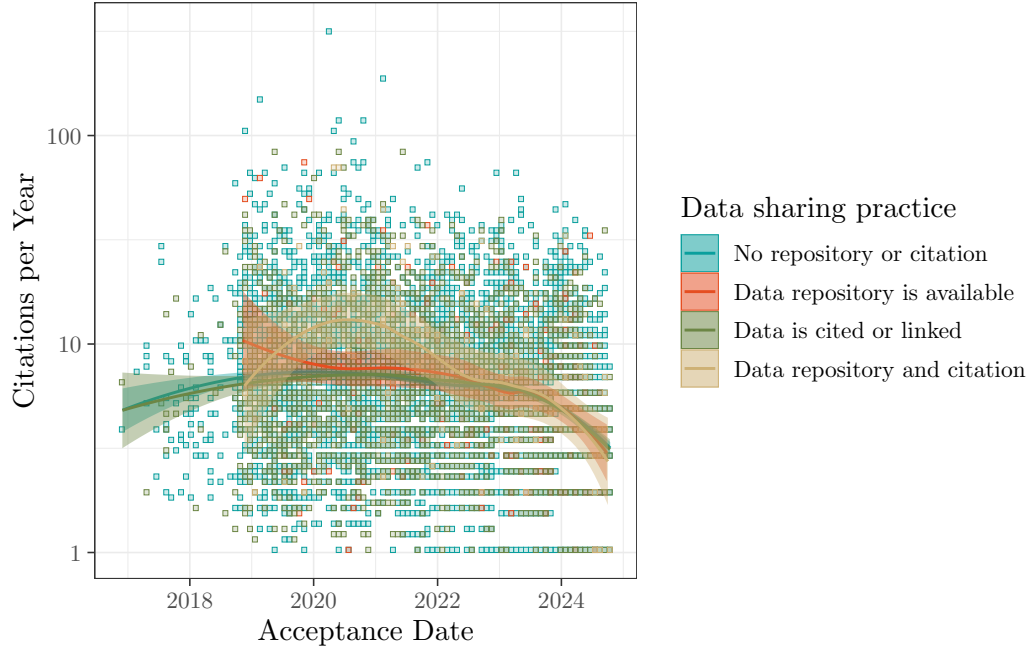


Figure 1: Citations per year of each paper over time by code and data-sharing practice, with a LOESS moving average line applied for mean comparison.

```

"lda_topicSocial & Policy Aspects of Mobility" = "Topic: Social & Policy Aspects of Mobility",
"lda_topicElectric Vehicles & Ride-Sharing" = "Topic: Electric Vehicles & Ride-Sharing",
"lda_topicRoad Infrastructure & Emergency Management" = "Topic: Road Infrastructure & Emergency Man
"region_normalizedSouth America" = "Region: South America"
)

modelsummary(
  list(
    "Controls" = lm_reviewtime_ctl,
    "Open Science" = lm_reviewtime,
    "Controls" = lm_citations_ctl,
    "Open Science" = lm_citations),
  estimate = "{estimate}{stars} ",
  statistic = "[{conf.low}, {conf.high}]",
  coef_map = cm,
  gof_omit = "AIC|BIC|F",
  notes = c("95% confidence interval in parentheses",
    "Full controls: Journal, paper topic, and region of corresponding author.",
    "Only significant controls displayed."),
  format = "tinytable"
) |>
  tinytable::group_tt( j = list( "Review time [days]" = 2:3, "Citations per year" = 4:5))

```

Table 2 presents estimates of log-linear regression models of the review time in days and citations per year for the papers in the dataset. These models include the journal, paper topic, and region of corresponding author as dummy controls to account for differing practices between journals and reviewers in topics. Insignificant parameter estimates on these controls are suppressed in Table 2 for clarity but the controls by themselves explain virtually all of the reduction in variance in the models, with insignificant reductions in the model F -statistic when markers for data sharing and code availability are included (p -value of 0.44 for review time and 0.33 for citations per year).

There are many considerations in this analysis, and in contemplating the role of open science practices vis a vis researcher incentives. First, researchers might not necessarily expect to see reduced review times with open science practices. If a reviewer can follow and reproduce the analysis in the paper during the review, this may result in more thorough reviews and more extended resubmission processes. Improved and

Table 2: Incentives for Open Science

	Review time [days]		Citations per year	
	Controls	Open Science	Controls	Open Science
Intercept	5.189*** [5.010, 5.369]	5.198*** [5.019, 5.378]	1.262*** [1.022, 1.503]	1.262*** [1.022, 1.503]
Code is available		0.013 [−0.051, 0.076]		−0.001 [−0.063, 0.061]
Data repository is available (NC, R)		−0.018 [−0.099, 0.064]		−0.001 [−0.063, 0.061]
Data is cited or linked (C, NR)		−0.019 [−0.045, 0.007]		−0.001 [−0.063, 0.061]
Data repository and citation (C, R)		0.052 [−0.068, 0.171]		−0.001 [−0.063, 0.061]
Journal: TR - A	0.755*** [0.710, 0.800]	0.754*** [0.709, 0.800]	0.415*** [0.353, 0.476]	0.415*** [0.353, 0.476]
Journal: TR - B	0.619*** [0.563, 0.675]	0.616*** [0.560, 0.672]	0.523*** [0.446, 0.601]	0.523*** [0.446, 0.601]
Journal: TR - C	0.415*** [0.368, 0.462]	0.413*** [0.366, 0.460]	0.722*** [0.656, 0.787]	0.722*** [0.656, 0.787]
Journal: TR - D	0.033 [−0.017, 0.083]	0.032 [−0.018, 0.081]	0.600*** [0.538, 0.661]	0.600*** [0.538, 0.661]
Journal: TR - E	0.305*** [0.253, 0.356]	0.303*** [0.252, 0.355]	0.682*** [0.610, 0.753]	0.682*** [0.610, 0.753]
Journal: TR - F	0.223*** [0.171, 0.274]	0.220*** [0.168, 0.271]	0.231*** [0.159, 0.303]	0.231*** [0.159, 0.303]
Topic: COVID-19 Impact on Travel & Activities	−0.213** [−0.347, −0.080]	−0.213** [−0.347, −0.080]	0.020 [−0.160, 0.200]	0.020 [−0.160, 0.200]
Topic: Social & Policy Aspects of Mobility	−0.097 [−0.225, 0.030]	−0.100 [−0.228, 0.028]	0.265** [0.091, 0.438]	0.265** [0.091, 0.438]
Topic: Electric Vehicles & Ride-Sharing	−0.035 [−0.170, 0.100]	−0.035 [−0.170, 0.100]	0.206* [0.025, 0.388]	0.206* [0.025, 0.388]
Topic: Road Infrastructure & Emergency Management	0.197+ [−0.019, 0.412]	0.195+ [−0.021, 0.411]	−0.270* [−0.537, −0.003]	−0.270* [−0.537, −0.003]
Region: South America	0.047 [−0.105, 0.199]	0.044 [−0.108, 0.197]	−0.209* [−0.409, −0.009]	−0.209* [−0.409, −0.009]
Num.Obs.	9185	9185	10 480	10 480
R ²	0.182	0.183	0.113	0.113
R ² Adj.	0.180	0.180	0.111	0.111
Log.Lik.	−7138.461	−7136.564	−11 825.901	−11 825.901
RMSE	0.53	0.53	0.75	0.75

95% confidence interval in parentheses

Only significant controls displayed.

more comprehensive reviews would benefit science, but would not strengthen researcher incentives. Similarly, sharing data and code may lead to more citations from others who pursue subsequent research, but this effect is only likely if the original research question lends itself to continued study, and if the code and data are packaged in such a way that continued study is feasible. Further, the models presented in Table 2 only explain one-tenth to one-fifth of the variance in the data: important variables such as individual reviewer availability, the complexity of the specific research question and methods, and other sources of variance are not available to this analysis but likely play a large role.

References

Nosek, B.A., Alter, G., Banks, G.C., Borsboom, D., Bowman, S.D., Breckler, S.J., Buck, S., Chambers, C.D., Chin, G., Christensen, G., et al., 2015. Promoting an open research culture. *Science* 348, 1422–1425.